Raffles Institution Physics Department Revision Package 1	Temperature and Ideal Gases	12

Important note: ****denotes questions that may require knowledge from later chapters in order to solve them.***

I. MCQs

1. **[Dunman High/H2/Prelims 2010/P1/16]**

   Which of the following statements about thermodynamic (Kelvin) scale is **not** correct?

   A Conversion between Celsius scale to Kelvin scale is $T/°C = T/K + 273.15$.

   B At zero Kelvin, the total internal energy of the system is at a minimum.

   C Absolute zero cannot be reached due to laws of thermodynamics.

   D The Kelvin scale does not depend on any thermometric properties.

2. **[Dunman High/H2/Prelims 2010/P1/17]**

   Material X of temperature T_X and material Y of temperature T_Y are in contact with each other until there is no net transfer of energy between X and Y. Which of the following statements is most *correct*?

   A X and Y may not be in thermal equilibrium with each other.

   B The internal energy of X is equal to the internal energy of Y.

   C The specific heat capacity of X must be equal to that of Y.

   D The average kinetic energies of material X must be equal to that of Y.

3. **[HCI/H2/Prelims 2010/P1/16]**

   Container X contains neon gas and container Y contains argon gas. Container X has twice the volume of container Y. The temperatures of the gases in both containers are the same. What is the ratio of the mean kinetic energy of a neon molecule to the mean kinetic energy of an argon molecule?

   [The relative atomic masses of neon and argon are 20 and 40 respectively.]

   A 0.5 B 1 C 2 D 4

4. **[IJC/H2/Prelims 2010/P1/16]**

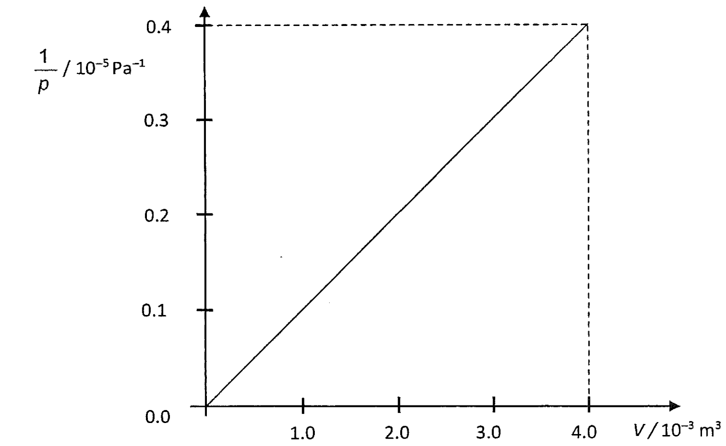
   The temperature of a body at 100°C is increased by $\Delta\theta$ as measured on the Celsius scale.

   How is this temperature change expressed on the Kelvin scale?

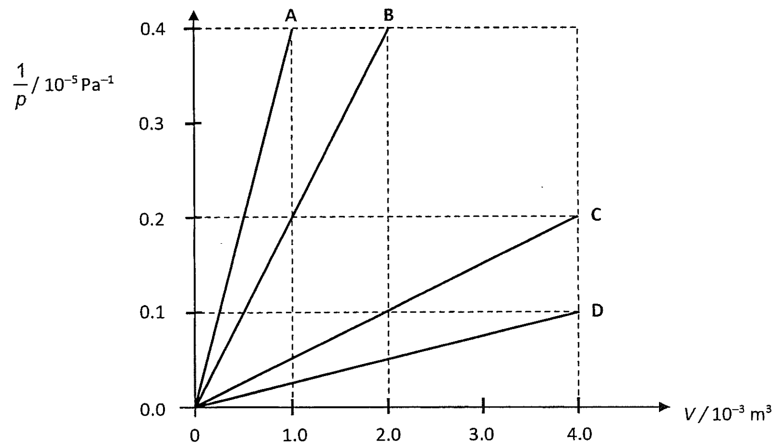
   A $\Delta\theta$ B $\Delta\theta + 273$ C $\Delta\theta + 273.15$ D $\Delta\theta + 273.16$

5. **[HCI/H2/Prelims 2010/P1/15]**

   n moles of an ideal gas has pressure p and volume V. The graph shows how $\frac{1}{p}$ varies with V at a constant temperature.



   If the number of moles of the gas is increased to $2n$ and the thermodynamic temperature is reduced to one quarter of the initial temperature, which graph will be obtained?



6. **[MJC/H2/Prelims 2010/P1/22]**

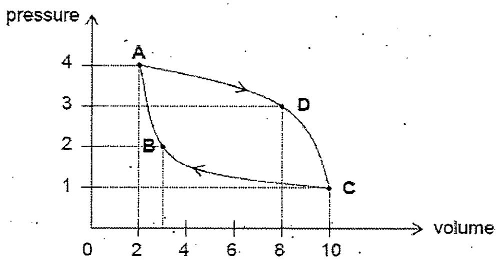
   An ideal gas is contained in a cylinder with a movable piston. At pressure p, volume V and temperature T, it has N_v molecules per unit volume. If the pressure of the gas is changed to $0.50p$, and the temperature to $2.0T$, the number of molecules per unit volume becomes

   A $0.25\,N_v$ B $0.50\,N_v$ C $1.0\,N_v$ D $4.0\,N_v$

7. [NJC/H2/Prelims 2010/P1/21]

A fixed mass of ideal gas undergoes a cycle of changes as shown in the figure.

At which point on the graph is the gas hottest?



8. [NYJC/H2/Prelims 2010/P1/17]

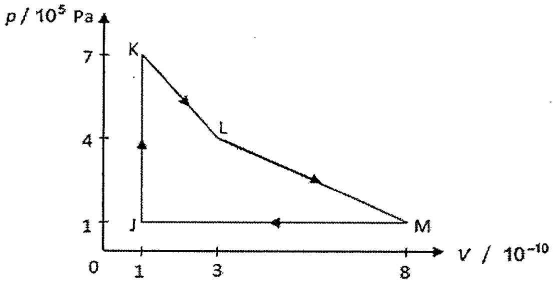
For an ideal gas at 300°C, the mean kinetic energy of the gas molecules is 1.6×10^{-23} J.

What are the values for the temperature of the gas and the mean kinetic energy of the gas molecules when its molecules are, on average, travelling at twice the speed?

	temperature / °C	kinetic energy / J
A	600	3.2×10^{-23}
B	1100	3.2×10^{-23}
C	2000	6.4×10^{-23}
D	2300	6.4×10^{-23}

9. [PJC/H2/Prelims 2010/P1/18]

A mass of an ideal gas of volume V and pressure p undergoes the cycle of changes shown in the graph below.



Which of the following shows the states of the gas arranged in the order of decreasing temperature?

A JKLM B JKML C KLMJ D LMKJ

10. [RI/H2/Prelims 2010/P1/18]*

The piston of a gas-tight syringe containing an ideal gas is pulled outwards quickly.

Which of the following changes is **not** correct?

A The density of the ideal gas decreases.

B The pressure of the ideal gas decreases.

C The temperature of the ideal gas decreases.

D The root-mean-square speed of the atoms increases.

11. [RVHS/H2/Prelims 2010/P1/17]

Two ideal gases X and Y, are contained in a cylinder at constant temperature. The mass of the atoms of X is m and of Y is $4m$.

Which one of the following is the correct value of the ratio $\dfrac{\text{average kinetic energy of atoms X}}{\text{average kinetic energy of atoms Y}}$?

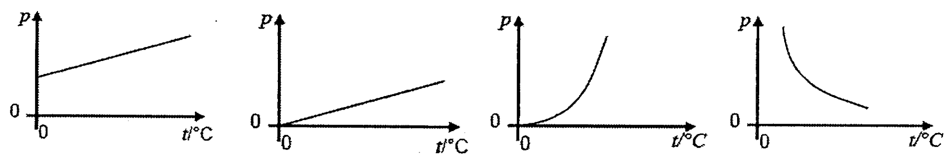
A 1 B 2 C 4 D 16

12. [SAJC/H2/Prelims 2010/P1/15]

A fixed mass of an ideal gas is heated at constant volume.

Which one of the following graphs best shows the variation with Celsius temperature t of pressure p of the gas?

A B C D



13. [TJC/H2/Prelims 2010/P1/19]

An inexpansible vessel initially contains 0.020 kg of gas at 300 K.

What is the mass of gas that has escaped from the vessel if it is heated from 300 K to 400 K under constant pressure?

A 0.0050 kg B 0.0075 kg C 0.012 kg D 0.015 kg

14. [VJC/H2/Prelims 2010/P1/18]

An ideal gas is held in a container of volume V at pressure P. The root-mean-squared (r.m.s.) speed of a gas molecule under these conditions is u.

If the volume and pressure are now changed to $2V$ and $2P$, the r.m.s. speed of a molecule will be

A u B $\sqrt{2}u$ C $2u$ D $4u$

15. [YJC/H2/Prelims 2010/P1/17]

Which one of the following is essential for the equation $pV = nRT$ to be obeyed by a real gas?

A Changes should be at constant temperature.

B Changes should take place at constant volume.

C Pressures should be low.

D Temperatures must be higher than 273.15 K.

II. Structured Questions

1. [MI/H2/Prelims 2010/P2/3 (part)]

a. An ideal gas at constant pressure has its volume directly proportional to its absolute temperature.
Calculate the absolute temperature T when an ideal gas has volume 0.00825 m^3, assuming that the same mass of the ideal gas at the same pressure has volume 0.00424 m^3 at a temperature of 273 K. [2]

b. State the conversion formula from the Celsius scale (°C) to the thermodynamic absolute scale (K). [1]

[Total: 3]

2. [MJC/H2/Prelims 2010/P3/5]

The temperature T of an ideal gas at pressure p is defined by the equation:

$$p = nkT.$$

a. Identify the quantity n. [1]

b. State an equation relating k, the Boltzmann constant, and R, the universal gas constant. [1]

[Total: 2]

3. [NYJC/H2/Prelims 2010/P3/7 (modified)]

a. State what is meant by saying a temperature is on an *absolute* scale. [1]

[Total: 1]

4. [PJC/H2/Prelims 2010/P3/6]*

a. State what is meant by saying that a temperature is on an *absolute scale*. [1]

b. Explain what is meant by

i. an *ideal gas*; [1]

ii. *absolute zero* on the Kelvin scale; [1]

iii. the *internal energy* of a gas. [2]

[Total: 5]

5. [NJC/H2/Prelims 2010/P3/2]

A monoatomic ideal gas is subject to a cycle of changes **ABCA**. Fig. 5.1 shows a graph of pressure p against volume V for one cycle of changes for the gas.

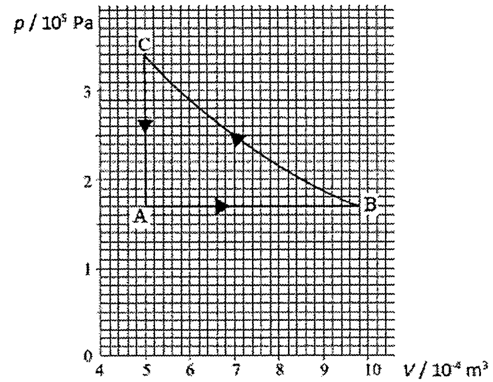


Fig. 5.1

Use the Kinetic Theory of gases to explain why the pressure of an ideal gas increases in the change **BC** when it contracts at constant temperature. [2]

[Total: 2]

6. [SAJC/H2/Prelims 2010/P3/6]

a. Fig. 6.1 represents how the temperature of a small mass of water changes when it is heated steadily from room temperature to above its boiling point in a large sealed container.

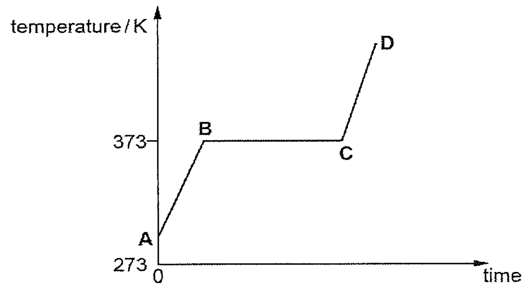


Fig. 6.1

Describe and explain the features of the graph in terms of the changes which occur to the separation of the molecules and to their potential and kinetic energies. Three distinct sections of the graph have been labelled to aid your description

i. **A** to **B**; [2]

ii. **B** to **C**; [2]

iii. **C** to **D**. [2]

b. This question is about the atmosphere treated as an ideal gas.

i. The equation of state of an ideal gas is $pV = nRT$. Data about gases are often given in terms of density ρ rather than volume V.

Show that the equation of state for a gas can be written as

$$p = \frac{\rho RT}{M},$$

where M is the mass of one mole of gas **[2]**

ii. One simple model of the atmosphere assumes that air behaves as an ideal gas at a constant temperature. Using this model the pressure p of the atmosphere at a temperature of 20°C varies with height h above the Earth's surface as shown in Fig. 6.2.

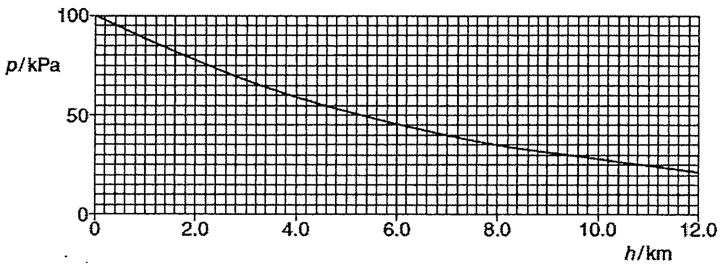


Fig. 6.2

Use data from the graph to show that the variation of pressure with height follows an *exponential* relationship. **[3]**

iii. The ideal gas equation in **b.i.** shows that, at constant temperature, pressure p is proportional to density ρ. Use data from Fig. 6.2 to find the density of the atmosphere at a height of 8.0 km.
(Density ρ of air at sea level = 1.3 kg m^{-3}) **[2]**

iv. In the real atmosphere the density, pressure and temperature all decrease with height. At the summit of Mt. Everest, 8.0 km above sea level, the pressure is only 0.30 of that at sea level. Take the temperature at the summit to be −23°C and at sea level to be 20°C.
Calculate, using the ideal gas equation, the density of the air at the summit.
(Density ρ of air at sea level = 1.3 kg m^{-3}) **[2]**

[Total: 15]

7. [TJC/H2/Prelims 2010/P2/4]*

A scuba diver releases an air bubble, of diameter 3.0 cm, from a depth of 14 m below the sea level. Assume that air behaves as an ideal gas and the temperature of water is constant at 25°C.

a. Given that the density of water is 1000 kg m^{-3} and the atmospheric pressure is 1.0×10^5 Pa, show that the pressure of the water at a depth of 14 m is 2.4×10^5 Pa. **[1]**

b. Hence calculate the volume of the air bubble when it reaches the surface of water. **[3]**

c. Sketch a clearly labelled graph in Fig. 7.1 showing the variation of pressure P with volume V of the air bubble as it rises from the sea. **[2]**

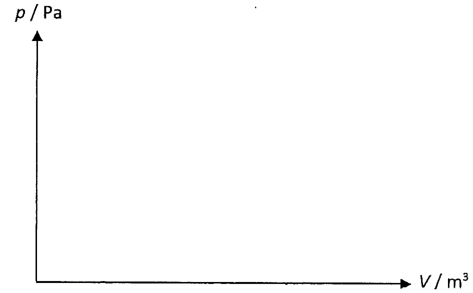


Fig. 7.1

d. Using the first Law of thermodynamics, deduce whether heat is added or removed from the air bubble as the bubble rises. **[2]**

[Total: 8]

	Raffles Institution Physics Department Revision Package 1	**Temperature and Ideal Gases**	**12**

III. Solutions and Answers

The suggested solutions and answers below are only for your reference and have not been verified to be error-free. If you have any doubt, please discuss with your tutor to clarify.

MCQs

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
A	D	B	A	B	A	D	C	D	D	A	A	A	C	C

1 A The correct conversion between Celsius scale to Kelvin scale is $T/°C = T/K - 273.15$.

2 D When 2 objects have the same temperature (i.e. particles in X and Y have the same average kinetic energy), there is no net transfer of thermal energy between them (i.e. the 2 objects are in thermal equilibrium).

3 B When 2 systems have the same temperature, the particles in both systems have the same average kinetic energy.

4 A The increment of 1°C is the same as the increment of 1 K.

5 B $PV = nRT$

$$\frac{1}{P} = (V)\left(\frac{1}{nRT}\right)$$

After the changes,

$$\frac{1}{P} = (V)\left(\frac{1}{2nR\left(\frac{T}{4}\right)}\right) = (2V)\left(\frac{1}{nRT}\right)$$

Thus the gradient of the graph is doubled.

6 A $pV = nRT$

$$N_v = \frac{n}{V} = \frac{p}{RT}$$

After the changes,

$$\frac{0.50p}{2RT} = \left(\frac{1}{4}\right)\left(\frac{p}{RT}\right) = 0.25N_v$$

7 D When temperature increase, isotherms on P-V graphs shift rightwards (outwards away from origin).

8 C $T \propto$ average KE of molecules (temperature in degree Kelvin)

Since speed is doubled, KE of molecules is quadrupled.

Temperature in degree Kelvin is also quadrupled

$(300 + 273) \times 4 = 2292$ K

$2292 - 273 = 2019$ °C

9 D $pV = nRT$

$pV \propto T$

$J \to pV = 1 \times 10^{-5}$

$K \to pV = 7 \times 10^{-5}$

$L \to pV = 12 \times 10^{-5}$

$M \to pV = 8 \times 10^{-5}$

10 D Note: knowledge of 1st Law of Thermodynamics is required to answer this question

According to 1st Law, when gas expands quickly, the gas is doing positive work and internal energy of gas decreases. Thus option D is false.

11 A When 2 systems have the same temperature, the particles in both systems have the same average kinetic energy.

12 A $p = \frac{nRT}{V}$ temperature in degree Kelvin

$p = \frac{nR(t + 273)}{V}$ temperature in degree Celsius

Thus p-t graph is translated leftwards when the units of temperature used is degree Celsius.

13 A Since vessel is inexpansible, volume of container is constant.

Mass of gas is directly proportional number of moles of gas.

$pV = nRT$

since p, V and R are constants, $n \propto \frac{1}{T}$

300 K $\to$ 400 K temperature increased 1.33 times

mass of gas left $= \frac{0.02}{1.33} = 0.01504$

mass of gas escaped $= 0.02 - 0.01504 = 4.96 \times 10^{-3}$ kg

14 C $PV = nRT$

$(2P)(2V) = 4nRT$

Since temperature is quadrupled, average KE of molecules is quadrupled and thus r.m.s of gas molecules is doubled.

15 **C** Pressure should be low so that the volume of molecules is negligible compared to the volume of container.

Structured Questions

1. **[MI/H2/Prelims 2010/P2/3]**

   a. $V \propto T \Rightarrow \dfrac{V_1}{V_2} = \dfrac{T_1}{T_2} \Rightarrow \dfrac{0.00825}{0.00424} = \dfrac{T_1}{273} \Rightarrow T_1 = 531\text{ K}$

   b. $T/\text{K} = \theta/°\text{C} + 273.15$

2. **[MJC/H2/Prelims 2010/P3/5]**

   a. n is the number density / number of molecules per unit volume

   b. $k = \dfrac{R}{N_A}$

3. **[NYJC/H2/Prelims 2010/P3/7 (modified)]**

   a. An absolute scale means that the scale is independent of the property of any particular substance.

4. **[PJC/H2/Prelims 2010/P3/6]***

   a. An absolute scale means that the scale is independent of the property of any particular substance.

   b. i. See definitions in lecture notes.

      ii. See definitions in lecture notes.

      iii. See definitions in lecture notes.

5. **[NJC/H2/Prelims 2010/P3/2]**

   Pressure of the gas is due to the collisions between the molecules and the container's walls.
   At a constant temperature, the molecules move at and collide with the same speed. However, due to the contraction in volume, the frequency of collisions increases.
   This implies that the rate of change of momentum due to the collisions increases.
   Hence, the force that the molecules exert on the walls increases.
   Thus, pressure increases.

6. **[SAJC/H2/Prelims 2010/P3/6]**

   a. i. **A** to **B**: kinetic energy increases with temperature.
   Separation, and thus potential energy, remains constant or increases very slightly.

      ii. **B** to **C**: kinetic energy remains constant as temperature remains constant.
   Separation, and thus potential energy, increases greatly, as phase transition occurs.

      iii. **C** to **D**: kinetic energy increases with temperature.
   Separation, and thus potential energy, remains constant or increases very slightly.

   b. i. $pV = nRT$

   $pV = \dfrac{m}{M}RT$

   $p = \dfrac{m}{V}\dfrac{RT}{M}$

   $p = \dfrac{\rho RT}{M}$ (shown)

      ii. Suitable test, i.e. ratio test, half-height, etc. (e.g. every 5.2 km, the pressure falls by about 50%)

      iii. From graph, at 8 km, $p = (3.5 \pm 0.3) \times 10^4$ Pa

   $\dfrac{p}{p_0} = \dfrac{\rho}{\rho_0} \Rightarrow \rho = \dfrac{p}{p_0}\rho_0 = \dfrac{3.5 \times 10^4}{1.01 \times 10^5} \times (1.3) = 0.45\text{ kg m}^{-3}$

      iv. $p = \dfrac{\rho RT}{M} \Rightarrow \dfrac{p}{\rho T} = \text{constant}$

   $\dfrac{p_0}{(1.3) \times (293)} = \dfrac{0.30 p_0}{\rho \times (250)}$

   $\rho = 0.45708 = 0.46\text{ kg m}^{-3}$

7. **[TJC/H2/Prelims 2010/P2/4]**

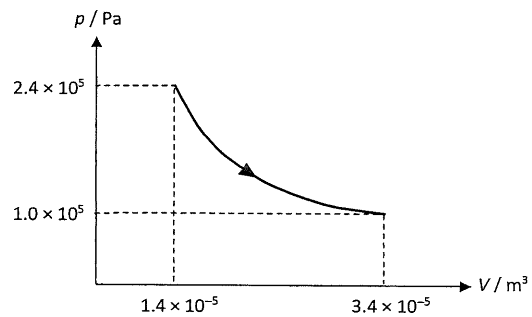
   a. $p = p_{atm} + \rho g h = (1.0 \times 10^5) + (1000)(9.81)(14) = 2.4 \times 10^5$ Pa (shown)

   b. Using the ideal gas equation for a constant temperature process,

   $p_i V_i = p_f V_f$

   $V_f = \dfrac{p_i V_i}{p_f} = \dfrac{(2.4 \times 10^5) \times \frac{4}{3}\pi \times (0.015)^3}{1.0 \times 10^5} = 3.39292 \times 10^{-5} = 3.4 \times 10^{-5}\text{ m}^3$

   c.



   d. At constant temperature, $\Delta U = 0$ for an ideal gas.
   Since volume of air bubble increases, work done on air bubble is negative.
   Using first Law of thermodynamics, Q is positive.
   Hence, heat is absorbed by the air bubble as its rises.